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SQA INTERVIEW Q/A

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SQA Interview Questions

1. What is Quality Assurance and Quality Control?

Quality Assurance (QA): Focuses on improving the processes to deliver quality products, ensuring that quality is built into the process. Quality Control (QC): Involves the actual testing of the product to ensure it meets the required standards and identifying defects.

2. What is testing?

Testing: The process of evaluating a system or its components to find whether it satisfies specified requirements, identifying defects, and ensuring the product is defect-free.

3. Software Testing Lifecycle?

Software Testing Lifecycle (STLC): Involves phases like requirement analysis, test planning, test case development, environment setup, test execution, defect reporting, and test closure.

4. Software Development Lifecycle?

Software Development Lifecycle (SDLC): A process that includes phases like requirement analysis, design, implementation, testing, deployment, and maintenance for developing software applications.

5. Difference between manual and automated testing?

Manual Testing: Performed by humans without using automation tools, suitable for exploratory, usability, and ad-hoc testing. Automated Testing: Uses scripts and tools to perform tests, ideal for regression, performance, and repetitive tasks.

6. Software Development Models?

Waterfall, Agile, V-Model, Spiral: These are methodologies used to structure, plan, and control the process of developing an information system.

7. Agile Testing Methodologies?

Agile Testing: Involves continuous testing of software in parallel with development, embracing methodologies like Scrum, Kanban, and Extreme Programming (XP).

8. Agile and Scrum?

Agile: A flexible, iterative approach to software development. Scrum: A framework within Agile for managing work, focusing on sprints and iterative progress through daily stand-ups and reviews.

9. Database Testing?

Database Testing: Involves verifying the integrity, accuracy, and consistency of data in databases, including testing of schema, tables, triggers, and procedures.

10. UI Testing?

UI Testing: Ensures that the graphical user interface meets specifications, checking elements like buttons, icons, and fields for functionality and visual correctness.

11. UX Testing?

UX Testing: Evaluates the user experience by assessing ease of use, efficiency, and satisfaction provided by the software to the end-users.

12. Usability Testing?

Usability Testing: A technique used to evaluate a product by testing it on users, focusing on ease of use, task completion, and user satisfaction.

13. API Testing?

API Testing: Involves testing application programming interfaces (APIs) directly to ensure they meet functionality, reliability, performance, and security expectations.

14. What is Test Ware?

Test Ware: All the materials produced during the testing process, including test cases, test plans, test scripts, and test data.

15. What is Test Artifact?

Test Artifact: Any document or work product created during the testing process, such as test plans, test cases, defect reports, and test scripts.

16. Why testing is important?

Importance of Testing: Ensures software quality, verifies functionality, identifies defects, reduces development costs, and increases user satisfaction and reliability.

17. Different Types of Testing?

Types of Testing: Includes unit testing, integration testing, system testing, acceptance testing, regression testing, performance testing, and usability testing.

18. Smoke Testing?

Smoke Testing: A preliminary test to check the basic functionality of an application, ensuring that the most critical features work before further testing.

19. Regression Testing?

Regression Testing: Re-tests software after changes to ensure existing functionalities are unaffected by new code or updates.

20. Unit Testing?

Unit Testing: Tests individual components or modules of software to ensure they work as expected, typically performed by developers during development.

21. What is Monkey Testing?

Monkey Testing: Random testing performed without any predefined test cases or plans to identify unexpected behaviors or crashes.

22. Ad-hoc Testing?

Ad-hoc Testing: Informal, unstructured testing performed without planning or documentation, aimed at finding defects through random checking.

23. Integration Testing?

Integration Testing: Tests the interfaces and interaction between integrated units/modules to ensure they work together correctly.

24. Sanity Testing?

Sanity Testing: A quick check to ensure that a particular function or bug fix works correctly after changes, performed before regression testing.

25. Retesting?

Retesting: The process of testing the same functionality again after a defect has been fixed to ensure the issue is resolved.

26. Functional Testing?

Functional Testing: Validates the software against the functional requirements, ensuring each function operates according to the specified behavior.

27. ELT Testing?

ELT Testing: Extract, Load, Transform testing involves verifying data extraction from sources, loading into the destination, and transforming it correctly within data warehouses.

28. What is Test Driven Development?

Test Driven Development (TDD): A software development approach where test cases are written before writing the code, driving the design of the software.

29. What is Data Driven Testing?

Data Driven Testing: An automation testing framework where test data is driven from external data sources like spreadsheets, databases, or CSV files.

30. Nonfunctional Testing?

Nonfunctional Testing: Evaluates aspects like performance, usability, reliability, and security, ensuring the software meets nonfunctional requirements.

31. What is Performance Testing?

Performance Testing: Measures the speed, responsiveness, and stability of software under various conditions to ensure it performs well under expected workloads.

32. Techniques of Testing?

Testing Techniques: Includes black box testing, white box testing, exploratory testing, ad-hoc testing, and risk- based testing.

33. BVA Approach?

Boundary Value Analysis (BVA): A testing technique focusing on the boundaries between partitions, identifying defects at the edge values.

34. ECP Approach?

Equivalence Class Partitioning (ECP): Divides input data into equivalence classes where all members are expected to be treated the same, reducing the number of test cases.

35. What is Validation?

Validation: Ensures the product meets the user's needs and requirements, verifying the final product through testing and user feedback.

36. What is Verification?

Verification: Ensures the product is being built correctly according to the specifications and design documents, typically through reviews and inspections.

37. Bug Lifecycle?

Bug Lifecycle: Includes stages like new, assigned, open, fixed, retest, verified, and closed, tracking the bug from identification to resolution.

38. What is Test Plan?

Test Plan: A document outlining the scope, approach, resources, and schedule of testing activities, defining what will be tested and how.

39. What is Test Scenario?

Test Scenario: A high-level description of what to test, including the possible actions and outcomes, based on user requirements and functionality.

40. What is Test Case?

Test Case: A detailed set of instructions for testing a particular feature or functionality, including preconditions, steps, inputs, expected results, and postconditions.

41. What is Use Case Testing?

Use Case Testing: Testing based on use cases that describe interactions between users and the system to ensure all possible scenarios are covered.

42. What is Test Script?

Test Script: A set of instructions written in a programming or scripting language to automate the execution of test cases.

43. What is JMeter?

JMeter: An open-source tool used for performance and load testing, simulating multiple users to test the performance of web applications.

44. What is Jira?

Jira: A popular project management tool used for issue tracking, bug tracking, and project management in Agile development environments.

45. What are Selenium and its flavors and their working?

Selenium: An open-source tool for automating web browser interactions. Flavors include Selenium WebDriver (cross-browser automation), Selenium IDE (record and playback), and Selenium Grid (parallel execution).

46. What is Mobile Application Testing?

Mobile Application Testing: Ensures the functionality, usability, and performance of mobile applications across different devices and operating systems.

47. SQL Queries?

SQL Queries: Commands used to interact with databases, performing operations like data retrieval (SELECT), insertion (INSERT), updating (UPDATE), and deletion (DELETE).

48. OOP Concepts?

OOP Concepts: Includes principles like encapsulation (data hiding), inheritance (parent-child relationship), polymorphism (many forms), and abstraction (simplified complexity).

49. Database Concepts, Stored Procedure?

Database Concepts: Involve understanding tables, relationships, normalization, and SQL operations. A stored procedure is a precompiled set of SQL statements stored in the database.

50. What are Cookies and its Types?

Cookies: Small pieces of data stored by web browsers to remember user information. Types include session cookies (temporary) and persistent cookies (stored on disk).

51. How to Test the Cookies?

Testing Cookies: Verify creation, expiration, data storage, and security by checking their behavior under different conditions and ensuring compliance with privacy policies.

52. Bug Priority and Severity?

Bug Priority: Indicates the urgency of fixing a bug. Bug Severity: Indicates the impact of the bug on the system's functionality.

53. Bug with High Priority and Low Severity?

Example: A typo in the company's homepage title is not severe but needs immediate fixing due to its high visibility.

54. Versions of Android, iOS and Web Browsers?

Versions: Regularly updated, with each platform releasing new versions periodically to add features and fix bugs (e.g., Android 12, iOS 15, Chrome 91).

55. Web Elements Spacing, Android, iOS?

Web Elements Spacing: Ensures elements are appropriately spaced for usability. For Android and iOS, guidelines specify minimum touch targets and spacing for accessibility.

56. What is Captcha and Why Use Captcha?

Captcha: A challenge-response test used to determine if the user is human, preventing automated bots from abusing online services.

57. Difference Between DBMS and RDBMS?

DBMS: Database Management System that manages databases. RDBMS: Relational Database Management System that uses a structured format and relationships between tables to manage data.

58. What is Penetration Testing?

Penetration Testing: A security testing technique to identify vulnerabilities by simulating attacks on the system.

59. What is SQL Injection?

SQL Injection: A code injection technique that exploits vulnerabilities in an application's software by inserting malicious SQL statements.

60. What is Brute Force Attack?

Brute Force Attack: A trial-and-error method used to decode encrypted data such as passwords by systematically trying all possible combinations.

61. What are Normalization and its Types?

Normalization: The process of organizing data in a database to reduce redundancy and improve data integrity. Types include 1NF, 2NF, 3NF, BCNF, etc.

62. What is Traceability Matrix (RTM)?

Traceability Matrix: A document that maps and traces user requirements with test cases, ensuring all requirements are covered by tests.

63. What is Clustering?

Clustering: A technique in data analysis where data is grouped into clusters that share similar characteristics, used for pattern recognition and classification.

64. Principles of Testing?

Principles of Testing: Includes early testing, defect clustering, pesticide paradox, exhaustive testing is impossible, testing shows presence of defects, and absence-of-errors fallacy.

65. What is Backlog or Product Backlog?

Backlog/Product Backlog: A prioritized list of features, enhancements, and bug fixes that are yet to be worked on in an Agile project.

66. Mobile App Performance Testing Checklist?

Checklist: Includes testing for load time, responsiveness, battery consumption, memory usage, network usage, and compatibility across devices and OS versions.

67. User Acceptance Testing?

User Acceptance Testing (UAT): The final phase of testing where actual users test the system to verify it can handle required tasks in real-world scenarios.

68. Fault, Failure and Error?

Fault: A defect in the system. Failure: When the system does not perform as expected due to a fault. Error: A human mistake causing a fault.

69. Exploratory Testing?

Exploratory Testing: Simultaneous learning, test design, and execution, emphasizing tester creativity and intuition.

70. Security Testing?

Security Testing: Ensures the software is free from vulnerabilities, protecting data and resources from threats and attacks.

71. What is Bug, Defect and Error?

Bug: A flaw in the software causing incorrect results. Defect: A deviation from the requirements. Error: A mistake made by a human.

72. What are Mutants and Mutation Testing?

Mutants: Small changes made to software code to create variations. Mutation Testing: A method to evaluate test cases by introducing faults (mutants) and checking if tests can detect them.

73. Basic Knowledge of HTML, CSS, JavaScript, Asp.net, OOP and Java?

HTML/CSS/JavaScript: Core web technologies for creating and styling web pages. ASP.NET: A framework for building web applications and services with .NET. OOP and Java: Object-oriented programming principles used in Java for modular, reusable code.

74. Entry and Exit Criteria?

Entry Criteria: Conditions that must be met before testing begins (e.g., test environment setup). Exit Criteria:

Conditions that must be met before testing is concluded (e.g., all critical bugs fixed).

75. Static & Dynamic Testing?

Static Testing: Reviews and inspections of code or documentation without executing the program. Dynamic Testing: Involves executing code and validating the output with expected results.

SQA Real-time Interview Questions

1. Write a java program for the largest number from three numbers

java

Copy code

```
public class LargestNumber {
public static void main(String[] args) {
        int a = 10, b = 20, c = 30;
        int largest = (a > b) ? (a > c ? a : c) : (b > c ? b : c);
        System.out.println("Largest number is: " + largest);
    }
}
```

2. What is SDLC and STLC? And Explain its phases

SDLC (Software Development Life Cycle): A process for planning, creating, testing, and deploying an information system, involving phases like requirement analysis, design, implementation, testing, deployment, and maintenance.

STLC (Software Testing Life Cycle): A sequence of specific activities conducted during the testing process to ensure software quality, involving phases like requirement analysis, test planning, test case development, environment setup, test execution, and test cycle closure.

3. Define your roles and responsibility.

Roles and Responsibilities: Responsibilities typically include analyzing requirements, creating and executing test cases, identifying defects, collaborating with development teams, ensuring product quality, and maintaining documentation.

4. What is regression testing?

Regression Testing: A type of software testing that ensures that recent code changes have not adversely affected existing functionalities of the software.

5. What are different methodologies of SDLC? Explain each

Waterfall: Sequential design process where progress is seen as flowing steadily downwards through phases. Agile: Iterative approach focusing on collaboration, customer feedback, and small, rapid releases. V-Model: Extension of the waterfall model where each development phase is associated with a testing phase. Spiral: Combines iterative development (prototyping) and systematic aspects of the waterfall model.

6. Define Agile.

Agile: A methodology that promotes continuous iteration of development and testing throughout the software development lifecycle, encouraging collaborative and flexible responses to change.

7. Define Scrum and Sprint.

Scrum: A framework within Agile for managing and completing complex projects, typically through incremental work called sprints. Sprint: A set time period within which specific work has to be completed and made ready for review.

8. What is the estimation in Sprint?

Sprint Estimation: The process of predicting the effort required to complete the tasks in a sprint, often using story points, hours, or other units of measure.

9. What is sprint backlog?

Sprint Backlog: A list of tasks and user stories selected from the product backlog to be completed during a sprint.

10. What are the different reports in Testing?

Testing Reports: Common types include Test Summary Report, Defect Report, Test Execution Report, and Traceability Matrix.

11. What are the key components of the Test Case report?

Test Case Report Components: Test case ID, description, steps, expected result, actual result, status (pass/fail), and remarks.

12. What are the components of a defect report?

Defect Report Components: Defect ID, summary, description, steps to reproduce, severity, priority, environment, assigned to, status, and attachments/screenshots.

13. What is Jira?

Jira: A popular project management tool used for bug tracking, issue tracking, and project management in Agile development.

14. How do you log a defect in Jira?

Log a Defect in Jira: Navigate to the appropriate project, click on "Create," fill in the defect details (summary, description, severity, priority, steps to reproduce), and save.

15. How do you link bugs with the user story?

Link Bugs with User Story: In Jira, use the "Link" option to associate the bug with the relevant user story, selecting the appropriate relationship (e.g., "relates to," "blocks," etc.).

16. What is sprint?

Sprint: A time-boxed iteration of continuous development in Scrum, typically lasting 1-4 weeks, aimed at delivering a usable product increment.

17. Define black box and white box testing.

Black Box Testing: Testing based on external expectations, without knowledge of internal code structure. White Box Testing: Testing based on knowledge of internal code structure, focusing on code logic, paths, and conditions.

18. Define functional testing.

Functional Testing: A type of testing that validates the software system against the functional requirements/specifications, focusing on what the system does.

19. Define the OOP concept in Java.

OOP Concept in Java: Object-Oriented Programming principles include encapsulation, inheritance, polymorphism, and abstraction, promoting modular and reusable code.

20. Give me examples of OOP which you used in your framework.

Examples of OOP in Framework: Using inheritance for base test classes, encapsulation in page object models, polymorphism for test execution, and abstraction to define interfaces for various modules.

21. What is TestNG?

TestNG: A testing framework inspired by JUnit and NUnit, designed to simplify a broad range of testing needs, from unit to integration testing.

22. What is usability testing?

Usability Testing: A technique used to evaluate a product by testing it on users to improve its usability, ensuring it is easy to use and understand.

23. What are the steps for reporting the defect in Jira?

Steps for Reporting a Defect in Jira: Identify the issue, click "Create" in Jira, enter the defect details (summary, description, steps to reproduce, severity, etc.), and submit the defect.

24. Define Structure of Selenium.

Structure of Selenium: Comprises Selenium IDE (recording tool), Selenium WebDriver (browser automation), Selenium Grid (parallel execution), and various language bindings (Java, C#, Python, etc.).

25. How will you handle the dropdown in Selenium?

Handle Dropdown in Selenium: Use the Select class to interact with dropdown elements:

java

Copy code

```
Select dropdown = new Select(driver.findElement(By.id("dropdownId")));
dropdown.selectByVisibleText("Option");
```

26. Different types of wait in selenium? Explain each of them.

Implicit Wait: Sets a default wait time for the entire session. Explicit Wait: Waits for a specific condition to be met before proceeding. Fluent Wait: Waits for a specific condition, polling at regular intervals and ignoring specific exceptions.

27. Difference between hard and soft assertion?

Hard Assertion: Immediately stops test execution upon failure. Soft Assertion: Collects all errors and continues execution, reporting all failures at the end.

28. Why are we using "WebDriver driver = new ChromeDriver()"? Why can't we
write "RemoteDriver driver = new ChromeDriver()"?

WebDriver vs. RemoteWebDriver: WebDriver is the interface implemented by ChromeDriver, whereas RemoteWebDriver is a class used for remote execution, not typically instantiated directly for local testing.

29. Explain the different Annotation in TestNG.

TestNG Annotations:

- @Test: Marks a method as a test.
- @BeforeMethod and @AfterMethod: Run before and after each test method.
- @BeforeClass and @AfterClass: Run before and after all test methods in the current class.
- @BeforeSuite and @AfterSuite: Run before and after all tests in the suite.

30. Define Priority and Severity of the Bug.

Priority: Indicates the order in which a bug should be fixed. Severity: Indicates the impact of the bug on the system's functionality.

31. How to maximize the screen in Selenium?

Maximize Screen in Selenium:

java

Copy code

```
driver.manage().window().maximize();
```

32. What are the different closure reports?

Closure Reports: Include Test Closure Report, Project Closure Report, and Incident Closure Report, summarizing the activities and findings after testing phases or project completion.

33. What is the difference between a Test Plan and a Test Strategy document?

Test Plan: A document detailing the scope, approach, resources, and schedule of testing activities. Test Strategy: A high-level document outlining the testing approach, objectives, and general principles to be followed during testing.

34. Define Bug lifecycle of JIRA.

Bug Lifecycle in JIRA: New → Assigned → Open → Fixed → Retest → Reopen (if not fixed) → Verified → Closed.